

SANJAVAN GHODASARA

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EDUCATION

Stevens Institute of Technology

Master of Science in Machine Learning

Silver Oak College of Engineering

Bachelor of Technology in Information Technology

Hoboken, NJ

Expected May 2027

Gujarat, India

May 2025

RESEARCH & PUBLICATIONS

Graduate Research Assistant - Educational Artificial Intelligence

Prof. Tiffany Li

Sep 2025 - Present

Stevens Institute of Technology

• **First-author publication:** "Can Learners Navigate Imperfect Generative Pedagogical Chatbots? An Analysis of Chatbot Errors on Learning." *ACM @ 2026*

• Analyzed 500+ student-chatbot interactions across 3 universities designing comprehensive qualitative taxonomy achieving high inter-rater reliability; implemented unsupervised machine learning framework identifying 5 distinct behavioral profiles from 576 learning sessions

• Developed Latent Profile Analysis pipeline using Gaussian Mixture Models with scikit-learn for probabilistic clustering; applied rigorous statistical validation (ANOVA, Bayesian Information Criterion, Akaike Information Criterion) replacing arbitrary classification thresholds

PROFESSIONAL EXPERIENCE

Data Scientist - Artificial Intelligence/Machine Learning Network Automation

Jio Platforms Limited (450M+ subscribers)

Jan 2025 - May 2025

Navi Mumbai, India

• Architected end-to-end geospatial machine learning system leveraging transfer learning with VGG19 convolutional neural network for nationwide 5G/6G site planning, achieving 93% land classification accuracy while processing 500,000+ satellite tiles daily across 23 states serving 100M+ users

• Deployed production inference pipeline on Google Cloud Platform using Docker containerization and Kubernetes orchestration (1.6s latency); designed custom QGIS plugin with integrated model validation interface reducing manual geographic information system work by 70% (12,000+ engineering hours saved annually)

Artificial Intelligence/Machine Learning Team Lead

IBM India Software Lab

May 2024 - Sep 2024

Ahmedabad, India

• Led 6-person engineering team developing BERT and GPT-based healthcare conversational agent deployed on IBM Cloud serving 50,000+ daily medical queries across 15 hospitals with 92% user satisfaction rating and sub-2-second response times

• Architected full-stack platform integrating real-time natural language understanding for medical query processing; intelligent system successfully integrated into IBM Watson Health commercial product suite generating \$2M+ annual recurring revenue

Data Analyst - Government Infrastructure Analytics

IBM India Pvt Ltd

July 2024 - Sep 2024

Ahmedabad, India

• Engineered National Highways Authority of India (NHAI) road safety dashboards and predictive analytics models analyzing 100,000+ traffic incidents across 50+ highways, improving stakeholder decision-making accuracy by 35% and informing \$250M+ infrastructure investments

• Executed end-to-end data pipeline using Python, Tableau, SQL, and IBM Watson Studio extracting actionable insights from complex real-world transportation datasets

Junior Software Developer

LTIMindtree

Aug 2022 - Feb 2023

Mumbai, India

• Architected scalable microservices architecture using FastAPI framework supporting 10,000+ daily API requests; shipped cross-platform Flutter mobile applications achieving 4.5+ star ratings with 50,000+ downloads and 99.9% crash-free user sessions

KEY PROJECTS

Synovia - Surgical Outcome Predictor

1st Place HackPrinceton (120+ teams, 800+ participants) | Clinical pilots

2025

• Surgical outcome prediction system using physics-based finite element analysis and Google Gemini multimodal reasoning delivering real-time neurological deficit predictions in under 10 seconds vs 6+ hours traditional pre-operative mapping; addresses 36% complication rate affecting 180,000+ annual brain surgeries; conducting IRB-approved clinical validation at 2 teaching hospitals

Synaptix - AI-Powered Drug Repurposing

MLH Track Winner, Hacklytics 2026 (Georgia Tech, 600+ participants)

2026

• Drug repurposing platform identifying new therapeutic uses for FDA-approved drugs; type disease with no cure, receive drug candidates with biological reasoning; addresses years-long drug development timelines by finding new uses for existing approved compounds

GeneAI - Pharmacogenomics Platform

Top 7 HackIllinois HackVoyagers Shark Tank (800+ participants)

2026

• Pharmacogenomics platform predicting adverse drug reactions based on patient genetic profiles; type genes and drug, receive clinical recommendation in seconds based on individual DNA metabolism vs population averages; addresses 1.3M+ annual ER visits from wrong drug prescriptions

UltraBench - Memory Benchmarking

DevHouse SF | Adopted by Supermemory

2025

• Open-source benchmarking infrastructure for retrieval-augmented generation (RAG) memory providers featuring CLI dashboard with automated testing across 4 standardized benchmarks testing 500+ scenarios; adopted by Supermemory as evaluation standard, cited by Pinecone and Weaviate documentation, achieving 5,000+ weekly downloads

Alama - One-Touch Onboarding

DevHouse SF | Requested by Based Hardware (Omi)

2025

• Conversational onboarding platform leveraging Browser Use autonomous agents with Chrome DevTools Protocol and Touch ID biometric authentication, replacing 47 OAuth permission screens with 30-second setup flow and increasing completion rate from 10% to 80%+; technology requested for production deployment in Omi companion wearable

HONORS & LEADERSHIP

• Hackathon victories: 1st place HackPrinceton Healthcare Track (600+ participants), DevHouse SF Overall (Founder round, 10 selected from 500+ builders), NexHacks (VC speed-dating selected, 1,500+ participants), Track Winner at Hacklytics 2026, Rutgers Health Hackathon; runners-up, Easycode Harvard x MIT, Columbia representing top 5% of 600-1,200 participant fields

• Publishing researcher: First-author ACM Learning @ Scale 2026 conference paper | IEEE Natural Language Processing Research Committee Lead | TEDx SOCIET Conference Director managing 2,100+ attendees with \$50,000 budget | **Captain Gujarat State Football Team (2021 Cup of India national tournament)**

• Technology acquisitions & adoptions: Alama onboarding platform requested by Based Hardware (Omi wearable) | UltraBench benchmarking standard adopted by Supermemory | Lead founding team @ **Luna** - intelligent social venue discovery platform | Leading textile similarity search system for Surat & Ahmedabad textile manufacturers (India's largest textile hub) using Computer Vision to reduce 92M tons of annual fabric waste

TECHNICAL SKILLS

Languages: Python, TypeScript, C++, Java, SQL, JavaScript

Machine Learning & AI: PyTorch, TensorFlow, Hugging Face Transformers, BERT, GPT, CLIP, Gemini, scikit-learn

Specializations: Physics-informed ML, Natural Language Processing, Computer Vision, Geospatial Intelligence, Mathematical Optim (CP-SAT, MILP)

Frameworks & Tools: React, FastAPI, Django, Flutter, Node.js, Docker, Kubernetes, Leaflet.js

Cloud & Databases: Google Cloud Platform, AWS, Vector Databases (FAISS, pgvector), PostgreSQL, MongoDB, Redis